

Homework 4

Math 2031 — Professor Hamilton

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Section 5.3

Question 2

Statement: Prove that $n^2 - n$ is divisible by 2 for every integer n .

Proof:

Let $n \in \mathbb{Z}$. We want to show that $\forall n \in \mathbb{Z}, 2|n^2 - n$.

There are two possible cases.

A) n is odd, i.e, $n = 2k_1 + 1$

B) n is even, i.e, $n = 2k_2$

Case 1: Odd: $n = 2k_1 + 1$

Let n be an odd integer, thus $n = 2k_1 + 1$ for some integer k_1 .

We want to show that $2|n^2 - n$.

By assumption since n is odd we can rewrite our expression as:

$$(2k_1 + 1)^2 - (2k_1 + 1)$$

Which expands to:

$$4(k_1)^2 + 4k_1 + 1 - 2k_1 - 1$$

Simplifying:

$$4(k_1)^2 + 2k_1$$

Factoring out a 2:

$$2(2(k_1)^2 + k_1)$$

Thus if n is odd, $2|n^2 - n$.

Case 2: Even: $n = 2k_2$

Let n be an even integer, thus $n = 2k_2$ for some integer k_2 .

We want to show that $2|n^2 - n$.

By assumption $n = 2k_2$ thus we can rewrite our expression as:

$$(2k_2)^2 - (2k_2)$$

Which expands to:

$$4(k_2)^2 - 2k_2$$

Factoring out a 2:

$$2(2(k_2)^2 - k_2)$$

$$\therefore \forall n \in \mathbb{Z}, 2|n^2 - n.$$

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Question 3

Statement: Prove that if $n \in \mathbb{Z}$, then $n^2 + 2$ is not divisible by 4 for every integer n .

Proof by Contradiction:

Let $n \in \mathbb{Z}$.

We want to show that $4 \nmid n^2 + 2, \forall n \in \mathbb{Z}$.

By contradiction, suppose not. Suppose $4|n^2 + 2$.

Thus, $n^2 + 2 = 4k_1, k_1 \in \mathbb{Z}$.

Simplifying further:

$$n^2 = 4k_1 - 2.$$

Case 1: n is even.

Assume n is even. Thus $n = 2c, c \in \mathbb{Z}$.

Thus:

$$(2c)^2 = 4k_1 - 2.$$

Expanding:

$$4c^2 = 4k_1 - 2.$$

Factor a 2:

$$2(2c^2) = 2(2k_1 - 1).$$

Cancel the 2:

$$2c^2 = 2k_1 - 1.$$

By Theorem 1.4.4, we know any even integer squared is also even, however here we $2c$, an even integer by assumption, equal to an odd integer, thus in this case we have encountered a contradiction.

Case 2: n is odd.

Assume n is odd. Thus $n = 2d + 1, d \in \mathbb{Z}$.

Thus by substitution:

$$(2d + 1)^2 = 4k_2 - 2$$

$$4d^2 + 4d + 1 = 4k_2 - 2$$

Rearranging:

$$4d^2 + 4d = 4k_2 - 2 - 1 \text{ Factoring a 2:}$$

$$2\left(\underbrace{d^2 + 2d}_{\text{Integer}}\right) = 2\left(\underbrace{k_2 - 1}_{\text{Integer}}\right) - 1$$

Here we encounter yet another contradiction, as the LHS by definition is even and the RHS is odd, thus we have encountered a contradiction.

Therefore, since the negation of the original statement is false, the original statement must be true.

If $n \in \mathbb{Z}$, then $n^2 + 2$ is not divisible by 4 for every integer n .

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Question 6c

Statement: If $a|b$ and $b|c$, then $a|c$.

Let $a, b, c \in \mathbb{N}$ such that $b = a(k_1)$ and $c = b(k_2)$.

We want to show that $a|c$, i.e, c is an integer multiple of a .

By assumption since $b = a(k_1)$ we can rewrite c as:

$$c = a(k_1)(k_2)$$

Thus we have shown $a|c$.

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Question 10

(A) $a = 1840, b = 1518$

$$\gcd(a, b) = 46$$

$$1840 = 1518(1) + 322$$

$$1518 = 322(4) + 230$$

$$322 = 230(1) + 92$$

$$230 = 92(2) + 46$$

$$92 = 46(2) + 0$$

Linear Combination:

$$46 = 1518(17) - 1840(14)$$

(B) $a = 1001, b = 3465$

$$\gcd(a, b) = 77$$

$$1001 = 3465(0) + 1001$$

$$3465 = 1001(3) + 462$$

$$1001 = 462(2) + 77$$

$$462 = 77(6) + 0$$

Linear Combination:

$$77 = 7(1001) - 2(3465)$$

Question 12

Statement: Let $a, b, c \in \mathbb{Z}$. Suppose that $\gcd(a, c) = \gcd(b, c) = 1$. Prove that $\gcd(ab, c) = 1$

Proof:

Let $a, b, c \in \mathbb{Z}$ and assume $\gcd(a, c) = \gcd(b, c) = 1$.

We want to show that $\gcd(ab, c) = 1$.

By assumption, since $\gcd(a, c) = 1$ and $\gcd(b, c) = 1$ it must follow by Bezout's Identity:

$ax + yc = 1$ and $bv + cd = 1$ where $d, v, x, y \in \mathbb{Z}$

Multiplying 1 by 1 from both sides we get:

$$(ax + yc)(bv + cd) = 1$$

Expand:

$$axbv + axcd + ycbv + yc^2d = 1$$

Factor out ab and c from the LHS:

$$ab \left(\underbrace{xv}_{\text{Integer}} \right) + c \left(\underbrace{axd + ybv + ycd}_{\text{Integer}} \right) = 1$$

$\therefore$ by the converse of Bezout's Identity $\gcd(ab, c) = 1$ which is what we wanted to show.

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Question 13

Statement: Let $a, b, c \in \mathbb{Z}$. Suppose that $a|c$ and $b|c$ and that $\gcd(a, b) = 1$. Prove that $ab|c$.

Proof:

Assume $a, b, c \in \mathbb{Z}$ such that $a|c \wedge b|c \wedge \gcd(a, b) = 1$.

We want to show that $ab|c$.

Our assumptions give us that $c = a(k_1), c = b(k_2)$ and $ax + by = 1$ for some integers k_1, k_2, x, y .

Given:

$$1 = ax + by$$

Multiply both sides by c :

$$c = cax + cby$$

By assumption $c = ak_1 = bk_2$:

$$c = abxk_2 + abyk_1$$

Factoring ab :

$$c = ab(xk_2 + yk_1)$$

$\therefore ab|c$.

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Additional Problems

Question 1

Statement: Let a, b, c be integers. Prove that there exists integers r and s for which $c = ar + bs$ if and only if $\gcd(a, b)|c$.

Proof:**Part 1:** $c = ar + bs \rightarrow \gcd(a, b) | c$ Let a, b, c be integers such that $c = ar + bs$.Let $d = \gcd(a, b)$. We want to show that $d | c$.By definition $a = dk_1$ and $b = dk_2$ for some integers k_1, k_2 .

We can rewrite our assumption as:

$$c = dk_1r + dk_2s$$

Factoring out d :

$$c = d \left(\underbrace{k_1r + k_2s}_{\text{Integer}} \right)$$

$$\therefore d | c.$$

Part 2: $\gcd(a, b) | c \rightarrow c = ar + bs$ Let a, b, c be integers such that $\gcd(a, b) = d$ and $d | c$.We want to show that $c = ar + bs$.

By definition and Bezout's Identity:

$$c = dk, d = ax + by$$

Thus we can express c as:

$$c = k(ax + by) = a(kx) + b(ky)$$

Letting $r = kx$ and $s = ky$ we have:

$$c = ar + bs.$$

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Question 2**Statement:** Let a, b, c and d be integers. Prove that if $\gcd(a, b) = 1, d | ac$ and $d | bc$, then $d | c$.**Proof:**Let $a, b, c, d \in \mathbb{Z}$ such that $\gcd(a, b) = 1, d | ac \wedge d | bc$.We want to show that $d | c$.By definition: $ax + by = 1, ac = d(k_1)$, and $bc = d(k_2)$ for some $x, y, k_1, k_2 \in \mathbb{Z}$.Take $ax + by = 1$ and multiply by c :

$$cax + cby = c$$

We know that $ac = ca = d(k_1)$ and $cb = bc = d(k_2)$, thus we can substitute:

$$dk_1x + dk_2y = c$$

Factoring d :

$$d \left(\underbrace{k_1x + k_2y}_{\text{Integer}} \right) = c$$

$$\therefore d | c.$$

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